

Multi-Seed Robustness Study

Why we ran it, and what it shows

Companion note to the technical report

Why a multi-seed study?

The main pipeline is reported on one fixed train/test split (`RANDOM_SEED=42`) for exact reproducibility. But several phases are stochastic — the VAE, the diffusion model, the fairness-penalised MLP, and the DQN all depend on random initialisation, and the stratified split itself depends on the seed — so a single run cannot separate a genuine effect from seed luck. The clearest warning sign was Phase 8 (the in-processing fairness penalty): the demographic-parity gap at $\lambda = 1$ read 0.086 in the originally committed run, 0.057 when re-run under a newer software stack, and 0.083 as the ten-seed mean — a swing wide enough to flip a conclusion. We therefore re-ran the *entire* pipeline over ten seeds.

What we did

Each seed drives a fresh stratified 75/25 split *and* fresh model initialisation, so the study is effectively a ten-fold Monte-Carlo cross-validation that varies the evaluation cohort and the optimisation randomness together. The seed is injected through a single environment variable read by the shared module, so the phase code is byte-for-byte identical across runs; the `seed_42` run reproduces the canonical results exactly. For each headline comparison we report the mean $\pm$ std over the ten seeds and a paired two-sided Wilcoxon signed-rank test on the per-seed pairs ($p = 0.002$ is the smallest attainable value for $n = 10$). Reproduce with `python run_all.py --multiseed` (or `code/run_multiseed.py`); the raw per-seed outputs, `aggregate.json`, and `aggregate_summary.csv` live in this folder.

What we found

Effect (mean over 10 seeds)	mean A	mean B	<i>p</i>
Multimodal > tabular F1 (RF / XGB / MLP)	.84 / .85 / .84	.78 / .79 / .77	0.002
Uncertainty > random active learning (F1)	0.795	0.758	0.002
RL agent < uncertainty (F1)	0.761	0.795	0.002
Fair-MLP $\lambda=1$ cuts DP vs $\lambda=0$	0.083	0.211	0.002
Diffusion halves VAE std-mismatch (Δ_σ)	0.124	0.271	0.002
Diffusion vs VAE downstream F1	within ± 0.005		n.s.
SMOTE vs VAE downstream F1	within ± 0.01		n.s.

Robust (survive seed-and-split variation at $p = 0.002$): the multimodal text lift on all three models; uncertainty sampling beating random; the RL agent *underperforming* uncertainty (an honest negative result that the single seed had flattered); the in-processing penalty cutting demographic-parity disparity; and the diffusion model halving the distributional-spread gap.

Not robust (so we no longer claim them): no oversampler — VAE, diffusion, or classical SMOTE — gives a meaningful downstream-F1 advantage over the others on this 32%-minority data; the generative win is distributional fidelity, not F1. And the precise Phase-8 operating point is seed-sensitive: the penalty robustly reduces disparity but at a real ~ 1.6 -point accuracy cost, and in the

ten-seed mean $\lambda = 1$ is weakly *dominated* by $\lambda \approx 5$ — so λ should be chosen as a region against a target disparity ceiling, not fixed at one “optimal” value.

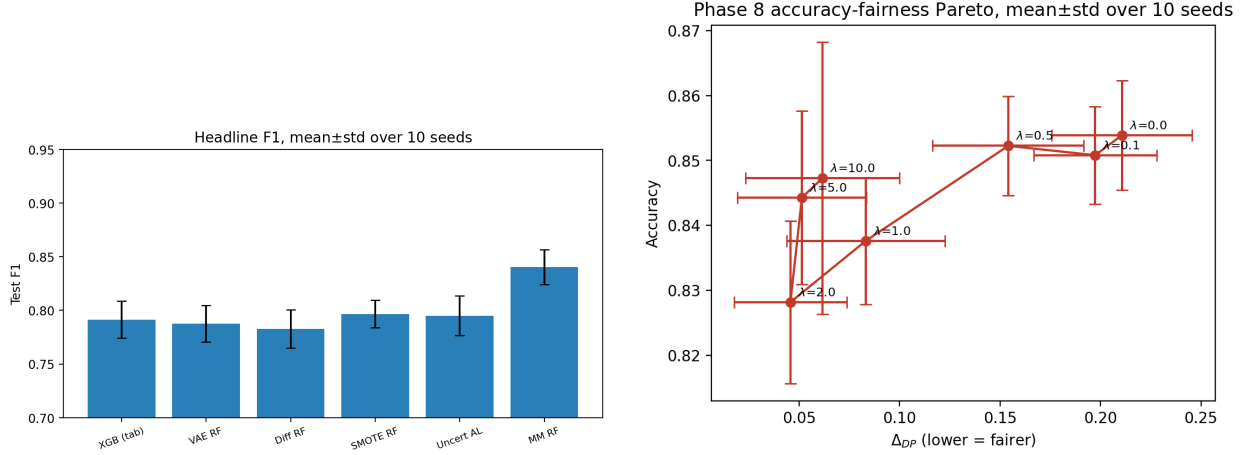


Figure 1: **Left:** headline test-F1 across the pipeline (mean $\pm$ std, 10 seeds) — the multimodal configurations clear the tabular baselines and the three oversamplers, which are mutually indistinguishable. **Right:** the Phase-8 accuracy/ Δ_{DP} Pareto frontier with across-seed error bars; $\lambda \approx 2$ –5, not $\lambda = 1$, traces the mean frontier.